

The State of Quantum AI

& AI for Quantum

A grounded briefing for technology executives: what is proven, what is emerging, and what is still hype — with citations to primary sources.

With reference to: IBM Think 2026 · NVIDIA Ising (Apr 2026) · Google DeepMind AlphaQubit (2024) · NIST PQC Standards (Aug 2024) · UCL Science Advances (Apr 2026) · McKinsey Quantum Technology Monitor 2025

SITUATION AT A GLANCE

Where the field actually stands

Quantum computing has entered its first commercially meaningful phase, moving from laboratory milestones toward narrow but real utility. The relationship between quantum and AI runs in **two directions: AI is being used to make quantum hardware more reliable, and quantum computing is beginning to augment classical AI** in specific high-value domains.

We are firmly in the Noisy Intermediate-Scale Quantum (NISQ) era. Fault-tolerant quantum computing, the kind that could break current encryption or run large-scale quantum AI, remains a research horizon. The most productive enterprise posture today is **hybrid**: classical systems doing the heavy lifting, with quantum used precisely where it adds measurable advantage.

\$750M+ Global quantum computing revenues in 2024; first \$1B crossing expected in 2025 <i>McKinsey Quantum Technology Monitor 2025</i>	12,000 Atoms in a molecule simulated by Cleveland Clinic + IBM quantum, April 2026 (up from 10 in Oct 2024) <i>IBM Think 2026</i>	2035 NIST deadline to deprecate quantum-vulnerable algorithms (RSA, ECC) from all federal standards <i>NIST IR 8547, 2024</i>
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DIRECTION 1

AI for Quantum

The most commercially advanced application of Quantum AI is not quantum computers running AI, but **classical AI stabilizing quantum hardware**. Qubits are fragile; any environmental disturbance causes decoherence and computational errors. AI is now the primary tool for managing this.

QUANTUM ERROR CORRECTION (QEC)

Google DeepMind's AlphaQubit (2024) is a transformer-based AI decoder that, in real time, outperforms hand-coded decoding algorithms on the Sycamore processor. It learns the unique noise fingerprint of its hardware and adapts continuously.

VERIFIED · APRIL 2026
NVIDIA Ising — the first open-source AI model family purpose-built for quantum processor calibration — was released **April 14, 2026** (not 2023 as sometimes cited). Ising Decoding delivers **2.5× faster** and **3× more accurate** error correction than pyMatching, the dominant open-source baseline. Ising Calibration reduces processor setup from days to hours. Early adopters include Harvard, Fermi National Lab, and Lawrence Berkeley National Laboratory.

AI-ASSISTED CHIP DESIGN

Deep learning "digital twin" models now simulate theoretical qubit layouts: predicting coherence times and crosstalk before physical fabrication begins. This is the most practical near-term entry point for classical ML engineers: analyzing hardware telemetry rather than writing quantum algorithms.

PRACTICAL IMPLICATION FOR ENTERPRISE
Your immediate quantum investment should not be buying hardware. Deploy existing senior ML engineering talent to understand quantum hardware telemetry and QEC. The first wave of commercialization is AI *operating* quantum systems.

DIRECTION 2

Quantum for AI

Quantum processors can augment classical ML in domains such as high-dimensional optimization and complex physical simulations. The architecture is consistently **hybrid**: classical systems handle data loading and general computation; quantum subroutines are inserted for specific tasks.

VARIATIONAL QUANTUM CIRCUITS (VQCS)

VQCs are the practical workhorse of today's quantum AI. A quantum processor runs a short, parameterized circuit; the results are fed to a classical optimizer (e.g., gradient descent), which updates the parameters and resubmits. This "shallow circuit" approach sidesteps decoherence by keeping quantum execution brief.

A real limitation: highly parameterized quantum circuits suffer from vanishing gradients: the **"barren plateau" problem**, where the optimization landscape becomes exponentially flat and training stalls. Engineers counter this with hardware-efficient circuit designs (ansätze) tailored to specific hardware noise profiles.

PROVEN USE CASES (AS OF MID-2026)

DOMAIN	WHAT QUANTUM ADDS	STATUS
Molecular simulation	IBM + Cleveland Clinic: 12,000-atom molecule simulation (Apr 2026), a ~1,200× scale leap from 10 atoms in Oct 2024	LIVE

Fluid dynamics / climate	UCL QIML framework: up to 17.25% accuracy gain and 29.36% fidelity gain over classical baselines, with ~100× less memory. Validated on turbulent flow equations (Science Advances, Apr 2026)	PUBLISHED
Combinatorial optimization	Portfolio optimization, logistics routing, energy grid planning. D-Wave annealers in production for discrete optimization workloads	PILOT / EARLY PROD.
EEG / biosignal classification	QSVM-QNN hybrid models (quEEGNet) achieve 0.990 classification accuracy on benchmark datasets, outperforming classical NNs under realistic noise conditions	RESEARCH
Financial market prediction	Theoretical advantage in stochastic, non-linear modeling. No peer-reviewed production evidence of quantum beating classical at scale as of this writing	CLAIMED , UNVERIFIED

THE DEQUANTIZATION CAVEAT — APPLY RUTHLESSLY

A purported "quantum speedup" often inspires a classical engineer to write a novel algorithm that matches the speed. If an advantage can be matched by a classical CPU, it is not true quantum utility. When evaluating vendor claims, require complexity-theory proof that the speedup scales in Hilbert space and cannot be replicated by a quantum-inspired classical algorithm on standard GPUs.

MOST URGENT NEAR-TERM ISSUE

Post-Quantum Security

This is not a future risk. It is a **present operational mandate**. Adversaries are today collecting encrypted data under a "harvest now, decrypt later" strategy: store ciphertext now, decrypt it when fault-tolerant quantum computers arrive. Data with a 10–15-year confidentiality requirement is already exposed.

NIST STANDARDS — FINALIZED AUGUST 2024

After an 8-year international evaluation process, NIST finalized its first three post-quantum cryptographic standards:

FIPS 203 ML-KEM Module-Lattice Key Encapsulation Mechanism (formerly CRYSTALS-Kyber). Primary replacement for RSA and ECDH key exchange. Comparatively small key sizes and fast operation.	FIPS 204 ML-DSA Module-Lattice Digital Signature Algorithm (formerly CRYSTALS-Dilithium). Primary standard for digital signatures, replacing ECDSA and RSA signatures.
FIPS 205 SLH-DSA Stateless Hash-Based Digital Signature Algorithm (formerly SPHINCS+). Backup signature scheme based on	March 2025 HQC (5th algorithm)

different mathematics — insurance if lattice vulnerabilities emerge.	Hamming Quasi-Cyclic, a code-based scheme added as a backup for ML-KEM key encapsulation, providing mathematical diversification.
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MIGRATION TIMELINE

Now	Begin inventory and migration planning NIST instructs organizations to migrate immediately. Identify every system that uses RSA, ECC, or classical Diffie-Hellman, and prioritize them by data sensitivity and lifespan.
2030	NSA CNSA 2.0 mandatory deadline All U.S. National Security Systems must complete migration to quantum-resistant cryptography. High-risk enterprise systems should be targeted for this date.
2035	NIST deprecation deadline Quantum-vulnerable algorithms (RSA, ECC) are removed from all NIST standards. Systems still using them will be out of compliance.

DO NOT WAIT FOR QUANTUM COMPUTERS TO ARRIVE
Migration of cryptographic infrastructure takes years, not weeks. The window to act is now, not when a fault-tolerant quantum computer makes headlines.

HARDWARE LANDSCAPE

Who is building what

No single quantum modality has won. Different architectures excel in different problem classes, and leadership is domain-specific rather than universal.

COMPANY	NOTABLE MILESTONE / DIFFERENTIATOR
IBM	Superconducting. Largest cloud quantum network; modular multi-chip architecture; 2026 quantum advantage target; transparent, metric-driven roadmaps that have consistently hit milestones
Google Quantum AI	Superconducting. 105-qubit Willow chip (late 2024) with scalable error correction; simulates 40-qubit systems using 1,000+ H100 GPUs in hybrid mode; AlphaQubit QEC
IonQ	Trapped ion. Focuses on algorithmic qubits and error performance rather than raw qubit count; strong gate fidelity metrics
Quantinuum	Trapped ion. Highest quantum volume benchmarks; emphasis on system fidelity; Microsoft partnership for topological qubit research

D-Wave

Quantum annealing. The only commercially deployed quantum computers producing enterprise revenue at scale today are purpose-built for discrete optimization problems

European Quantum Computer Companies (Hardware Focus)

As of May 2026, the European ecosystem has consolidated around several key hardware modalities, achieving record-breaking benchmarks and securing sovereign supply chains.

- **IQM Quantum Computers (Finland/Germany):** A leader in superconducting circuits; supplied the **Euro-Q-Exa** system to the Leibniz Supercomputing Centre (LRZ).
- **AQT - Alpine Quantum Technologies (Austria):** Specializes in trapped-ion quantum computing. In May 2026, AQT unveiled its **LYNX Series**, achieving a record-breaking **Quantum Volume (QV) of 32,768**, the highest for a European-built universal system.
- **PASQAL (France):** The primary provider for neutral atom-based quantum simulators, utilized in the **HPCQS** federated infrastructure.
- **Quandela (France):** A leader in photonic quantum computing; recognized with the **2026 Future Unicorn Award**. It has deployed five quantum computers across Europe and North America and launched Europe's first open-source quantum cloud.
- **Alice & Bob (France):** Focuses on universal fault-tolerant quantum computing using **Schrödinger cat qubits**, which implement autonomous quantum error correction.
- **QuantWare (Netherlands):** An industrial quantum processor provider that raised **\$178 million (€152 million)** in Series B funding in May 2026 to scale its **VIO-40K™** 10,000-qubit architecture.
- **QuantumCore Ltd (Germany):** A quantum infrastructure firm focused on superconducting amplifiers and cryogenic hardware, listed on the **Frankfurt Stock Exchange** in May 2026.

Quantum-as-a-Service (QaaS) & EuroHPC Stacks

The EuroHPC Joint Undertaking (JU) has established a world-class federated infrastructure that provides researchers and industry with seamless access to quantum-classical hybrid systems.

Key EuroHPC Quantum Stacks

- **Euro-Q-Exa (Germany):** Hosted at LRZ (Munich), this 54-qubit superconducting system (IQM Radiance technology) is integrated with the SuperMUC-NG supercomputer. A **150-qubit upgrade** is scheduled for late 2026.
- **LUMI-Q (Czechia):** Based on superconducting qubits with a **star-shaped topology** for one-to-all connectivity, hosted at the IT4Innovations National Supercomputing Center.
- **HPCQS (France/Germany):** A federated infrastructure integrating **100+ qubit neutral atom simulators** (PASQAL) into the Joliot-Curie (GENCI) and JUWELS (JSC) Tier-0 HPC systems.
- **Additional Sites:** Quantum systems are also integrated into **Leonardo** (Italy) and **MareNostrum 5** (Spain) to provide diverse hardware modalities (photonic, trapped-ion, etc.).

Access for Experimentation

- **EuroHPC Federation Platform (EFP):** Launched on **April 15, 2026**, this serves as a unified single access point for European researchers and businesses. It features a unified authentication and authorization (AAI) method to streamline access to both HPC and quantum resources.
 - **Munich Quantum Portal (MQP):** Provides localized access for users in Germany and across Europe to the Euro-Q-Exa system and hybrid workflows.
 - **Open-Source Integration:** Systems support industry-standard packages such as **Qiskit** and **PennyLane**, allowing for the execution of hybrid algorithms in logistics, finance, and molecular simulation.
 - **AI Factories:** 19 "AI Factories" across Europe now complement these stacks, offering customized support to SMEs and startups for developing quantum-informed AI models.
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EXECUTIVE ACTION PLAN

What to do — and in what order

- ▶ **Immediately: audit cryptographic exposure.** Inventory every system using RSA, ECC, or classical Diffie-Hellman. Prioritize by data sensitivity and lifespan. Systems holding data that must remain confidential for 10+ years are already at risk from harvest-now-decrypt-later attacks.
 - ▶ **Immediately: begin PQC migration planning.** Adopt NIST's finalized standards (ML-KEM, ML-DSA, SLH-DSA) as your target. Set internal deadlines that precede the 2030 NSA / 2035 NIST deprecation milestones.
 - ▶ **Near-term: upskill ML engineers on quantum hardware telemetry.** The first commercial wave of Quantum AI is AI managing quantum processors, not quantum processors running AI. Your senior ML talent can contribute via QEC and digital twin work without capital hardware purchases.
 - ▶ **Near-term: identify one high-value combinatorial bottleneck.** Portfolio optimization, molecular simulation, logistics routing, or energy grid planning. Pilot a hybrid VQC or quantum annealing model on that specific problem within 12 months. Do not generalize beyond it prematurely.
 - ▶ **Ongoing: evaluate vendor claims with the dequantization test.** If a vendor claims a quantum speedup, require proof that the advantage cannot be replicated by a quantum-inspired classical algorithm on standard hardware. Many claimed speedups fail this test.
 - ▶ **Ongoing: Use cloud access before buying hardware.** IBM Quantum, AWS Braket, Google Quantum AI, and Azure Quantum all offer access to cloud QPUs. Validate the actual advantage on your specific workloads before committing capital expenditure.
 - ▶ **Do not: form a committee to "study quantum."** Monitoring without piloting is the fastest path to obsolescence. Equally, avoid overcommitting resources to use cases: financial market prediction, autonomous SOC, that lack peer-reviewed production evidence at scale.
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SOURCES

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- NIST IR 8547; HQC standard March 2025 ·
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This briefing represents an automated snapshot with human supervision of the field as of May 2026. All market projections should be treated as indicative. This is not investment advice.